

Namespace Teradyne.ARCM.RA.FailedTests. AMPCommunication

Classes

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Provides utility definitions for the DAS system.

[DASListener](#)

Represents a DAS listener for managing incoming requests and events.

Delegates

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Delegate for handling new message events.

[DASListener.DasErrorHandler](#)

Delegate for handling errors during DAS operations.

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Delegate for DAS connection and disconnection events.

Class AMPEvents

Namespace: [Teradyne.ARCM.RA.FailedTests.AMPCommunication](#)

Assembly: FailedTests.dll

Provides utility definitions for the DAS system.

```
public class AMPEvents
```

Inheritance

[object](#)  ← AMPEvents

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Delegate AMPEvents.NewMessageDlg

Namespace: [Teradyne.ARCM.RA.FailedTests.AMPCommunication](#)

Assembly: FailedTests.dll

Delegate for handling new message events.

```
public delegate void AMPEvents.NewMessageDlg(string cellid, AMPMessageModel message)
```

Parameters

cellid [string](#)

The identifier of the test cell that received the message.

message [AMPMessageModel](#)

The [AMPMessageModel](#) instance representing the received message.

Class DASListener

Namespace: [Teradyne.ARCM.RA.FailedTests.AMPCommunication](#)

Assembly: FailedTests.dll

Represents a DAS listener for managing incoming requests and events.

```
public class DASListener : IDisposable
```

Inheritance

[object](#)  ← DASListener

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

DASListener(string)

Initializes a new instance of the [DASListener](#) class with the specified URL.

```
public DASListener(string dasur1)
```

Parameters

dasur1 [string](#) 

The URL for the DAS listener.

Fields

_messageList

List of messages managed by this application

```
protected AMP_MESSAGE_LIST _messageList
```

Field Value

[AMP_MESSAGE_LIST](#)

Properties

DASURL

Gets or sets the DAS URL for the listener.

```
public string DASURL { get; set; }
```

Property Value

[string](#)

Methods

Dispose()

Releases all resources used by the DASListener.

```
public void Dispose()
```

Start()

Starts the DAS listener asynchronously.

```
public Task<bool> Start()
```

Returns

[Task](#) [<bool>](#)

A task that represents the asynchronous operation. Returns true if the listener was started successfully; otherwise, false.

Exceptions

[HttpListenerException](#)

Thrown when the HTTP listener encounters an error.

Stop()

Stops the DAS listener and cleans up resources.

```
public void Stop()
```

subscribeToListOfMessages(List<string>, bool)

Allows DAS applications to specify the list of messages to subscribe to.

```
public void subscribeToListOfMessages(List<string> messageNames, bool resetFirst = true)
```

Parameters

messageNames [List](#) [<string>](#)

List of messages to subscribe to

resetFirst [bool](#)

Possibility to unsubscribe from all messages first (enabled by default)

Events

Connected

Event triggered when a connection is established.

`public event DASListener.DasEventHandler Connected`

Event Type

[DASListener.DasEventHandler](#)

Disconnected

Event triggered when the listener is disconnected.

`public event DASListener.DasEventHandler Disconnected`

Event Type

[DASListener.DasEventHandler](#)

Error

Event triggered when an error occurs.

`public event DASListener.DasErrorHandler Error`

Event Type

[DASListener.DasErrorHandler](#)

NewMessage

Event triggered when a new message is received.

`public event AMPEvents.NewMessageDlg NewMessage`

Event Type

[AMPEvents.NewMessageDlg](#)

Delegate DASListener.DasErrorHandler

Namespace: [Teradyne.ARCM.RA.FailedTests.AMPCommunication](#)

Assembly: FailedTests.dll

Delegate for handling errors during DAS operations.

```
public delegate void DASListener.DasErrorHandler(Exception exception)
```

Parameters

exception [Exception](#) 

The exception that occurred.

Delegate DASListener.DasEventHandler

Namespace: [Teradyne.ARCM.RA.FailedTests.AMPCommunication](#)

Assembly: FailedTests.dll

Delegate for DAS connection and disconnection events.

```
public delegate void DASListener.DasEventHandler()
```

Namespace Teradyne.ARCM.RA.FailedTests. AMPMessages

Classes

[AMPMessageModel](#)

Represents a AMP message model containing details about a message, its metadata, and test cell information.

[AMP_MESSAGE_LIST](#)

Managing the list of AMP Messages. It is mandatory to return a list of message names as a response to the INITIALIZATION message and recommended for each message.

Class AMPMessageModel

Namespace: [Teradyne.ARCM.RA.FailedTests.AMPMessages](#)

Assembly: FailedTests.dll

Represents a AMP message model containing details about a message, its metadata, and test cell information.

```
public class AMPMessageModel
```

Inheritance

[object](#)  ← AMPMessageModel

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Properties

BMessage

The formatted (pretty-printed) JSON message if valid, or the original message if not.

```
public string BMessage { get; }
```

Property Value

[string](#) 

Message

The raw message content in JSON format. When set, attempts to parse and format the message as indented JSON.

```
public string Message { get; set; }
```

Property Value

[string](#)

MessageName

The name of the message.

```
public string MessageName { get; set; }
```

Property Value

[string](#)

TestCellID

The identifier for the test cell associated with the message.

```
public string TestCellID { get; set; }
```

Property Value

[string](#)

TimeStamp

The timestamp of the message.

```
public DateTime TimeStamp { get; set; }
```

Property Value

[DateTime](#)

URL

The URL segments associated with the message.

```
public string[] URL { get; set; }
```

Property Value

[string](#)[↗][]

isInitialization

This is the INITIALIZATION message

```
public bool isInitialization { get; }
```

Property Value

[bool](#)[↗]

Class AMP_MESSAGE_LIST

Namespace: [Teradyne.ARCM.RA.FailedTests.AMPMessages](#)

Assembly: FailedTests.dll

Managing the list of AMP Messages. It is mandatory to return a list of message names as a response to the INITIALIZATION message and recommended for each message.

```
public class AMP_MESSAGE_LIST
```

Inheritance

[object](#) ← AMP_MESSAGE_LIST

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

AMP_MESSAGE_LIST()

Initialization. By default, all messages are selected.

```
public AMP_MESSAGE_LIST()
```

Fields

AMP_MESSAGES

All existing AMP messages

```
protected readonly List<string> AMP_MESSAGES
```

Field Value

[List](#) <[string](#)>

Properties

MessageList

Current list of messages used by the DAS application

```
public List<string> MessageList { get; }
```

Property Value

[List](#) <[string](#)>

Methods

ExtractMessageListFromInitMessage(AMPMessageModel)

The INITIALIZATION message sends the list of available messages.

```
public void ExtractMessageListFromInitMessage(AMPMessageModel message)
```

Parameters

message [AMPMessageModel](#)

INITIALIZATION message

ToJSON()

Creates a JSON string based on the list of message names currently subscribed to.

```
public string ToJSON()
```

Returns

[string](#)

the JSON string to send as a response

subscribe(List<string>)

Subscribe to a list of messages. Passing an empty list is not an error but has no effect.

```
public void subscribe(List<string> messageNames)
```

Parameters

messageNames [List](#) <[string](#)>

List of messages to subscribe to

subscribe(string)

Adds a message name to the list of messages we are interested in.

```
public bool subscribe(string messageName)
```

Parameters

messageName [string](#)

Name of the message to subscribe to

Returns

[bool](#)

Success or failure

subscribeAll()

We are interested in all existing messages.

```
public void subscribeAll()
```


unsubscribeAll()

Empty the list of messages we are interested in.

```
public void unsubscribeAll()
```

updateListToMatchAvailableMessages(AMP_MESSAGE_LIST)

Intersect lists to only keep the messages that are both available and of interest to the DAS application.

```
public void updateListToMatchAvailableMessages(AMP_MESSAGE_LIST available)
```

Parameters

available [AMP_MESSAGE_LIST](#)

List of available messages as provided by the INITIALIZATION message